

## AQM Problem Set 1 Homework #1

### Problem 1.1. Two-charge-carrier Drude model:

Consider a system of two types of charge carriers in the Drude model. The two carriers have the same density ( $n$ ) and opposite charge ( $e$  and  $-e$ ), and their masses and relaxation rates are  $m_1, m_2$  and  $\tau_1, \tau_2$ , respectively. So their corresponding mobilities are  $\mu_1 = \frac{e\tau_1}{m_1}$  and  $\mu_2 = \frac{e\tau_2}{m_2}$ , respectively.

- Calculate the magnetoresistance  $\Delta\rho = \rho(B) - \rho(B=0)$ , where  $B$  is the magnetic field.
- Calculate the Hall coefficient.
- In an undoped semiconductor,  $n = n_0 e^{-\Delta/k_B T}$  describes the temperature dependence of the carrier concentration. What will the temperature dependence of the magnetoresistance and the Hall coefficient be?

(a) for the two carries, we have

$$\sigma_{0i} = \frac{ne^2\tau_i}{m_i} = ne\mu_i, \quad \omega_{ci} = \frac{eB}{m_i}, \quad \omega_{ci}\tau_i = \mu_i B.$$

The total conductivity tensor is

$$\hat{\sigma} = \hat{\sigma}_1 + \hat{\sigma}_2$$

and the components are

$$\sigma_{xx} = ne \left( \frac{\mu_1}{1+(\mu_1 B)^2} + \frac{\mu_2}{1+(\mu_2 B)^2} \right), \quad \sigma_{xy} = B \cdot \sigma_{xx}, \quad \sigma_{zz} = ne(\mu_1 + \mu_2).$$

The resistance tensor is the inverse

$$\hat{\rho} = \hat{\sigma}^{-1} = \frac{1}{\sigma_{xx}^2 + \sigma_{xy}^2} \begin{bmatrix} \sigma_{xx} & -\sigma_{xy} \\ \sigma_{xy} & \sigma_{xx} \\ 0 & 0 & \sigma_{zz} \end{bmatrix}$$

Then, the element  $\rho_{xx}$  becomes.

$$\rho_{xx}(0) = \sigma_{xx}^{-1}(0) = \frac{1}{ne(\mu_1 + \mu_2)}, \quad \text{and} \quad \rho_{xx}(B) = \frac{\sigma(B)}{\sigma_{xx}^2(B) + \sigma_{xy}^2(B)}.$$

So, the magnetoresistance is

$$\Delta\rho = \rho_{xx}(B) - \rho_{xx}(0) = \frac{1}{ne} \left[ \frac{\frac{\mu_1}{1+(\mu_1 B)^2} + \frac{\mu_2}{1+(\mu_2 B)^2}}{\left( \frac{\mu_1}{1+(\mu_1 B)^2} + \frac{\mu_2}{1+(\mu_2 B)^2} \right)^2 + \left( \frac{\mu_1}{1+(\mu_1 B)^2} - \frac{\mu_2}{1+(\mu_2 B)^2} \right)^2 B^2} - \frac{1}{\mu_1 + \mu_2} \right]$$

(b). Since.

$$R_{xy}(B) = - \frac{\sigma_{xy}(B)}{\sigma_{xx}(B) + \sigma_{yy}(B)}$$

substitute it into  $R_H = \rho_{xy}/B$ . that is

$$R_H = - \frac{1}{ne} \left[ \frac{\frac{M_1}{1+(\mu_1 B)^2} + \frac{M_2}{1+(\mu_2 B)^2}}{\left( \frac{M_1}{1+(\mu_1 B)^2} + \frac{M_2}{1+(\mu_2 B)^2} \right)^2 + B^2 \left( \frac{M_1}{1+(\mu_1 B)^2} - \frac{M_2}{1+(\mu_2 B)^2} \right)} \right]$$

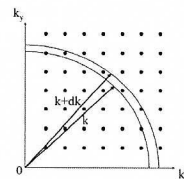
(c). For simplicity, consider the case  $B$  is small. now,  $\Delta p$  &  $R_H$  becomes

$$\Delta p = ne \mu_1 \mu_2 (\mu_1 + \mu_2) \left( \frac{\mu_1 - \mu_2}{\mu_1 + \mu_2} \right)^2 B^2, \quad R_H = - \frac{1}{ne} \frac{\mu_1 - \mu_2}{\mu_1 + \mu_2}$$

So, in this case,  $\Delta p \propto e^{-\Delta/k_B T}$ , while  $R_H \propto e^{+0/k_B T}$ .

**Problem 1.2.** Density of States in 2D system:

Consider free electron in a 2D box with a width of  $a$ . The allowed wave-numbers  $k_x$  and  $k_y$  for  $x$  and  $y$  directions respectively are expressed as  $\frac{\pi n_x}{a}$  and  $\frac{\pi n_y}{a}$ , where  $n_x, n_y = 1, 2, 3, \dots$  are integer numbers. The following figure shows the schematic view of the 2D array of allowed quantum states in  $k$ -space.



- Determine the area in  $k$ -space occupied by one allowed  $k$  point.
- Express number of allowed states in the area enclosed by the first quarter circles with the radius of  $k$  and  $k + dk$ , where  $k \gg dk$  (Consider the spin degeneracy).
- Express density of allowed states in real space per unit energy.

1a). The spaces of unit  $k$  point in  $k_x$ - and  $k_y$ - directions are

$$\Delta k_x = \frac{\pi}{a}, \quad \Delta k_y = \frac{\pi}{a}$$

So, the area in  $k$ -space occupied by one allowed  $k$  point is

$$\Delta k_x \cdot \Delta k_y = \left(\frac{\pi}{a}\right)^2$$

1b). The area enclosed by the two quarter circle is

$$dA = \frac{1}{4} \cdot 2\pi k \cdot dk = \frac{\pi k}{2} dk$$

Since every  $k$  point's area is  $\left(\frac{\pi}{a}\right)^2$ , so, the number of  $k$  points in this area is

$$dN = \frac{2dA}{\left(\frac{\pi}{a}\right)^2} = \frac{ka^2}{\pi} dk$$

Since two electrons can occupy each grid point ( $|\uparrow\rangle$  and  $|\downarrow\rangle$ ).

1c). Since,

$$dE = \frac{\hbar^2 k}{m} dk \rightarrow dk = \frac{m \cdot dE}{\hbar^2 k}$$

Substitute it into the results in (b):

$$dN = \frac{ka^2}{\pi} \cdot dk = \frac{ma^2}{\pi \hbar^2} dE$$

So, the density of allow states in real space per unit energy is

$$g(E) = \frac{1}{a^2} \cdot \frac{dN}{dE} = \frac{m}{\pi \hbar^2}$$

**Problem 1.3.** Density of states from a general formula:

- (a) In a 3D solid and starting from the general formula,  $g(E) = \frac{2V}{(2\pi)^3} \iint \frac{dS_E}{\nabla_k E}$ , find the expression for the electronic density of states: (i) for free electrons; (ii) for electrons that obey the isotropic dispersion relation:  $E = -E_0 - 2\gamma \cos\left(\frac{ka}{2}\right)$ , where  $E_0, \gamma > 0$ .
- (b) After having established the corresponding expression for 2D, find  $g(E)$  for (i) and (ii) above.
- (c) Same question for 1D.

Firstly, calculate sth. before since

$$E = -E_0 - 2\gamma \cos(ka/2)$$

we can obtain

$$\sin\left(\frac{ka}{2}\right) = \sqrt{1 - \left(\frac{E+E_0}{2\gamma}\right)^2}, \text{ and } k = \frac{2}{a} \cos^{-1}\left(-\frac{E+E_0}{2\gamma}\right)$$

as well as  $k = \sqrt{2mE}/\hbar$ ,  $\nabla_k E = \hbar^2 k/m$ , or  $\gamma a \sin(ka/2)$  for 1D.

(a). In 3D situation,

$$g(E) = \frac{2V}{(2\pi)^3} \cdot \frac{4\pi k^2}{\hbar^2 k/m} = \frac{Vm}{\pi^2 \hbar^3} \sqrt{2mE}$$

For 1D,

$$g(E) = \frac{V k^2}{\pi \gamma a \sqrt{1 - \left(\frac{E+E_0}{2\gamma}\right)^2}}$$

(b). In 2D situation,

$$g(E) = \frac{2A}{(2\pi)^2} \cdot \frac{2\pi k}{\hbar^2 k/m} = \frac{mA}{\pi \hbar^2} = \text{const.}$$

For 1D,

$$g(E) = \frac{2A}{(2\pi)^2} \cdot \frac{2\pi k}{\gamma a \sin(ka/2)} = \frac{2A}{\pi a^2} \frac{\cos^{-1}\left(-\frac{E+E_0}{2\gamma}\right)}{\sqrt{1 - \left(\frac{E+E_0}{2\gamma}\right)^2}}$$

(c). In 1D situation,  $g(E) = \frac{2L}{2\pi} \cdot \frac{2}{dE/dk}$ .

$$g(E) = \frac{2L}{\pi \cdot dE/dk} = \frac{L}{m\hbar} \sqrt{\frac{2m}{E}}$$

For 1D,

$$g(E) = \frac{2L}{\pi a} \frac{1}{\sqrt{1 - \left(\frac{E+E_0}{2\gamma}\right)^2}}$$